Four-zero tail-palindromic polynomials over $\mathbb{F}_2$: reciprocal obstructions and quadratic-transform intersections

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Abstract

We study monic even-degree polynomials over $\mathbb{F}_2$ whose nonleading coefficients form a palindrome with constant term one and exactly four zeros. We completely classify their intersections with full-degree quadratic rational transformations. The translated Q branch consists of two infinite shape families and three degree-14 exceptions; the translation-invariant branch consists of four reducible polynomials at degrees 14 and 30. More generally, a translation-invariant tail palindrome of even degree $n \geq 6$ has $n+2$ a power of two and is reducible whenever its number of missing tail coefficients is divisible by four. The classification determines the boundary of quadratic degree doubling: every four-zero output has degree 2 modulo 4, so iterated doubling from an even input degree cannot produce a member of the family. We also determine the exact reciprocal common divisor and count the position pairs surviving this obstruction. Reproducible arithmetic computations identify 2663 primitive polynomials at the 98 even degrees from 6 to 200 and verify the first solutions through degree 300. Conjecture C_4 on primitive existence at every even degree remains open.

1 The family and the question

Quadratic rational transformations are a standard tool for constructing irreducible polynomials of doubled degree [8]. This paper asks which outputs can also satisfy a rigid coefficient constraint: a palindromic tail with exactly four zero coefficients.

Put $n = 2k \geq 6$ and $S_n(x) = 1 + x + \cdots + x^n$. We consider

$$f_{n,i,j}(x) = S_n(x) + x^i + x^j + x^{n-1-i} + x^{n-1-j}, \qquad 1 \leq i < j \leq k-1, \tag{1}$$

over $\mathbb{F}_2$. These are exactly the monic degree-n polynomials with constant term one whose nonleading coefficient vector is a palindrome with four zero entries. The restrictions on i, j exclude the constant position. There are $\binom{k-1}{2}$ candidates. Let $N_4(n)$ denote the number of primitive polynomials in this family. Here primitive means irreducible with a root of multiplicative order $2^n - 1$; standard background is in [2].

Conjecture 1.1 (C_4)**.** For every even $n \geq 6$, one has $N_4(n) > 0$.

Primitive polynomials with prescribed coefficients are studied systematically by Cohen [1]. In the present family the field is fixed, all but four tail coefficients are one, and the four missing positions are coupled by reversal. The number of coefficient constraints therefore grows with the degree. A tail palindrome can also be encoded by the self-reciprocal polynomial $xf(x) + 1$ of odd degree $n + 1$. This describes its support; irreducibility and primitivity refer throughout to f itself.

The main results completely classify the full-degree quadratic outputs. Theorems 3.3 and 3.8 give the translated Q and translation-invariant lists. A trace obstruction also yields uniform reducibility when the number of missing tail coefficients is divisible by four (Proposition 3.7). Corollary 3.9 combines the branches and rules out iterated doubling from an even input degree. The proof converts the coefficient constraints into coupled translation identities, whose solutions are determined by binary submask arguments.

Section 2 gives a complementary arithmetic screen: an exact reciprocal common divisor and a closed count of the surviving position pairs (Propositions 2.2 and 2.4).

Section 4 gives reproducible primitive counts and first witnesses. Section 5 develops the questions suggested by the classification, while Appendix A describes coefficient multiplicities and finite-field root parameters.

2 The reciprocal obstruction

Tail symmetry turns polynomial reciprocity into an explicit common-factor test. In the four-zero family, the forced factor is determined by two integer gcds, and the number of surviving position pairs has a closed formula. The calculation uses the binomial gcd identity [2] and unit counts expressed by the second Schemmel function [5, Section 1].

Lemma 2.1 (Reciprocity and multiplicities)**.** *Let $f = x^n + \sum_{r=0}^{n-1} c_r x^r$, where n is even, $c_0 = 1$, and $c_r = c_{n-1-r}$. Write $f^*(x) = x^n f(1/x)$. Then*

$$f^* = xf + x^{n+1} + 1, \qquad \gcd(f, f^*) = \gcd(f, x^{n+1} + 1).$$

This gcd is squarefree. In particular, every self-reciprocal irreducible factor of f occurs with multiplicity one. If a non-self-reciprocal irreducible g and g^* both divide f, the smaller of their multiplicities is one.

Proof. The coefficient symmetry gives the displayed identity, hence the gcd identity by subtraction. The derivative of $x^{n+1} + 1$ is x^n, which is coprime to it; thus its divisors are squarefree. A self-reciprocal factor has the same multiplicity in f and f^*. For a reciprocal pair the two multiplicities are interchanged, so their minimum is the multiplicity in the gcd. This proves the claims. □

Proposition 2.2 (Exact reciprocal common divisor)**.** *For $f = f_{n,i,j}$ from (1), put*

$$N = n + 1, \quad \Delta = j - i, \quad R = n - 1 - i - j, \quad a = \gcd(N, \Delta), \quad b = \gcd(N, R), \quad h = \gcd(a, b).$$

Then, as an exact polynomial quotient in $\mathbb{F}_2[x]$,

$$\gcd(f, f^*) = \frac{(x^a + 1)(x^b + 1)}{(x + 1)(x^h + 1)}, \qquad \deg \gcd(f, f^*) = a + b - h - 1. \tag{2}$$

Consequently, irreducibility of f requires $a = b = 1$.

Proof. The nonleading terms of f occur in pairs, so $f(1) = 1$. By Lemma 2.1, with $S_n = (x^N + 1)/(x + 1)$,

$$\gcd(f, f^*) = \gcd(f, S_n) = \gcd(S_n, P), \qquad P = x^i(1 + x^\Delta)(1 + x^R).$$

The oddness of N makes S_n squarefree. For positive u, v, the Euclidean algorithm on the exponents gives $\gcd(x^u + 1, x^v + 1) = x^{\gcd(u,v)} + 1$ over $\mathbb{F}_2$. Applying this identity yields

$$\gcd(S_n, 1 + x^\Delta) = \frac{x^a + 1}{x + 1}, \qquad \gcd(S_n, 1 + x^R) = \frac{x^b + 1}{x + 1}.$$

Since S_n is squarefree, its gcd with their product is their least common multiple. Their intersection is $(x^h+1)/(x+1)$, proving (2). If this gcd were all of f, monicity and equal degrees would give $f = f^*$, hence $(x+1)f = x^N+1$ and $f = S_n$, contrary to its four zero coefficients. Thus a nonconstant gcd is a proper divisor. The right side is constant exactly when $a = b = 1$. □

Corollary 2.3 (An explicit infinite reducible family)**.** *For every odd $m \geq 3$, the valid position pair $(n,i,j) = (3m-1, m-2, m)$ satisfies*

$$f_{3m-1,m-2,m}(x) = (1 + x + \cdots + x^{m-1}) \cdot (x^{2m} + x^{m+1} + x^{m-1} + x^{m-2} + 1). \tag{3}$$

In particular it is reducible, and the displayed forced divisor has unbounded degree $m-1$.

Proof. The inequalities $1 \leq m-2 < m \leq (3m-3)/2$ make the positions valid. Here $N = 3m$, $\Delta = 2$, and $R = m$, so $a = 1$, $b = m$ in Proposition 2.2. For the explicit quotient, group the geometric sum as $S_{3m-1} = S_{m-1}(1 + x^m + x^{2m})$. The four deleted terms sum to

$$x^{m-2}(1+x^2)(1+x^m) = S_{m-1}x^{m-2}(x+1)^3.$$

Adding these expressions and cancelling the two x^m terms inside the second factor proves (3). □

Proposition 2.4 (Number of positions surviving the screen)**.** *For odd $N = n+1$, let φ be Euler's totient and put*

$$\mathfrak{S}_2(N) = N \prod_{p \mid N} \left(1 - \frac{2}{p}\right).$$

The number of four-zero position pairs with $\gcd(f, f^*) = 1$ is

$$E_N = \binom{\varphi(N)/2}{2} - \frac{\mathfrak{S}_2(N) - 1}{2}. \tag{4}$$

Thus $N_4(n) \leq E_{n+1}$. This is only an upper bound; $E_{n+1} > 0$ does not imply $N_4(n) > 0$.

Proof. Set $u = i+1$, $v = j+1$. The two coprimality conditions in Proposition 2.2 are precisely that $u - v$ and $u + v$ be units modulo N. Since 2 is invertible, the map $(u,v) \mapsto (u-v, u+v)$ is a bijection on $(\mathbb{Z}/N\mathbb{Z})^2$, giving $\varphi(N)^2$ ordered pairs before excluding zero coordinates. The cases $u = 0$ and $v = 0$ contribute $\varphi(N)$ each and are disjoint. The independent sign changes of u, v and their interchange act freely on the remaining pairs: admissibility excludes $u = \pm v$, and a nonzero coordinate cannot equal its negative. Dividing by eight gives $\varphi(N)(\varphi(N)-2)/8$ unordered pairs of distinct nonzero sign classes.

Choose representatives $1, \ldots, k$ of those classes. The allowed positions use $2, \ldots, k$, so the class represented by 1 must be removed. The Chinese remainder theorem counts $\mathfrak{S}_2(N)$ residues v for which both $v-1$ and $v+1$ are units: modulo each prime $p \mid N$ two residues are forbidden. One residue is $v = 0$; the others form pairs $\{v, -v\}$. Hence the removed class has $(\mathfrak{S}_2(N)-1)/2$ neighbors. Subtraction proves (4). □

Remark 2.5 (Factorization examples and fixed positions). For example,

$$f_{8,1,2} = (x^3 + x^2 + 1)(x^5 + x^2 + 1)$$

has trivial reciprocal gcd and is reducible. Repeated factors also occur:

$$f_{14,2,5} = (x^2 + x + 1)(x^6 + x^4 + x^3 + x + 1)^2.$$

The squarefreeness hypothesis in the classical Stickelberger–Swan factor-parity criterion [3] must therefore be checked. An odd number of factors still need not mean irreducibility: $f_{12,4,5}$ has three distinct factors of degrees $3, 4, 5$.

For fixed i, j, one also has

$$(x+1)f_{n,i,j} = x^{n-1-j}A_{i,j} + B_{i,j},$$

where $A_{i,j} = x^{j+2} + (x+1)(x^{j-i}+1)$ and $B_{i,j} = 1 + (x+1)(x^i + x^j)$ are fixed polynomials. This places fixed-position factor-parity questions in the classical framework of Fredricksen, Hales and Sweet [4].

3 Quadratic-transform intersections

3.1 The translated Q branch

For a degree-m polynomial h, define

$$\mathcal{T}(h)(x) = (x+1)^m h\left(\frac{x^2}{x+1}\right), \qquad \mathcal{M}_n(f)(x) = (x+1)^n f\left(\frac{x}{x+1}\right). \tag{5}$$

The map $\mathcal{T}$ is the classical Q-transform followed by the translation $x \mapsto x+1$; explicitly, $\mathcal{T}(h)(x) = h^Q(x+1)$. The equivalence between $\mathcal{M}_{2m}(f) = f$ and $f = \mathcal{T}(h)$ is a special case of Mattarei and Pizzato [8, Lemma 9]: take $a = b = 1, c = 0$, numerator x^2 and denominator $x+1$ in their notation. The nondegeneracy conditions hold, and the representation is unique; indeed the summands $x^{2j}(x+1)^{m-j}$ have distinct degrees $m+j$. More general invariant representations are given by Reis [9, Theorem 1.3]. We combine these invariant representations with the four-zero coefficient condition to determine every compatible output, including reducible polynomials.

Write $\Delta a = a(x+1) + a(x)$, and use reversal with respect to a specified formal degree when the leading coefficient may vanish.

Lemma 3.1 (A descent for tail palindromes)**.** *Let $f = x^{2k} + \sum_{j=0}^{2k-1} c_j x^j$, with $c_0 = 1$ and $c_j = c_{2k-1-j}$. Set $d = k-1$ and*

$$e = \sum_{a=0}^{k-1} c_{2a} x^a, \qquad o = x^d e(1/x).$$

Then

$$\mathcal{M}_{2k}(f) = f \quad \Longleftrightarrow \quad \Delta e = 0 \text{ and } e = \Delta(xo). \tag{6}$$

If the tail of f has $2s$ zeros, the formal coefficient vector of e has s zeros.

Proof. Tail symmetry gives $f = x^{2k} + e^2 + xo^2$. Put $p = xf + x^{2k+1} = xe^2 + x^2o^2$, and homogenize p to formal degree $2k+1$, obtaining $P(X,Y)$. The tail condition is exactly $P(X,Y) = P(Y,X)$. Moreover, $\mathcal{M}_{2k}(f) = f$ is equivalent to $P(X, X+Y) = P(X,Y)$. Conjugating this substitution by the interchange of X, Y gives $(X,Y) \mapsto (X+Y, Y)$. Hence, for this symmetric P, the condition is equivalent to $\Delta p = 0$. Direct expansion gives

$$\Delta p = x(\Delta e)^2 + \big(e(x+1) + \Delta(xo)\big)^2.$$

The two summands have disjoint odd and even supports. Thus both vanish exactly under the conditions in (6). Each tail mirror pair contains one even and one odd position, which proves the assertion about zeros. □

Lemma 3.2 (Two missing coefficients and translation)**.** *Let d be positive and even, and $1 \le a < b \le d-1$. Then $e = S_d + x^a + x^b$ satisfies $\Delta e = 0$ exactly in the following cases:*

	d	$\{a, b\}$
I	$2^r - 2,\ r \ge 3$	$\{2^u, 2^v\},\quad 0 \le u < v \le r-1$
II	$2^r,\ r \ge 2$	$\{2^r - 1, 2^t\},\quad 0 \le t \le r-1$
III	$2^r + 2^s - 2,\ r > s \ge 2$	$\{2^r - 1, 2^s - 1\}$.

(7)

Proof. A binary submask of an integer is obtained by retaining any subset of its nonzero binary digits; it is proper when it differs from the integer. The binary expansion identity $(x+1)^v = \prod_{j:\, v_j=1}(x^{2^j}+1)$ shows that Δx^v, for $v > 0$, consists of the proper binary submasks of v, and has $2^{\mathrm{wt}_2(v)} - 1$ terms. Consequently $\Delta x^v = 1$ exactly for powers of two. For distinct positive v, w, equality $\Delta x^v = \Delta x^w$ also occurs exactly when both are powers of two: with at least two binary digits, the singleton submasks recover all set digits. We will also use

$$x(x+1)\Delta S_t = x^{t+2} + (x+1)^{t+2} + 1. \tag{8}$$

It follows that $\Delta S_t = 0$ exactly when $t+2$ is a power of two.

For $0 \le b \le t$, the one-missing-coefficient polynomial $S_t + x^b$ is translation invariant exactly when

$$(t, b) = (2^s - 2, 0) \quad \text{or} \quad (2^s - 1, 2^s - 1), \qquad s \ge 1.$$

Indeed equality of its values at $0, 1$ forces $b = 0$ if t is even. If t is odd, translation invariance forces its leading term to cancel, since a nonzero invariant polynomial has even degree, so $b = t$. Equation (8) completes both cases, including the zero polynomial at $t = b = 0$.

Put $d = 2u$ and write $e = E^2 + xO^2$. Odd and even coefficient comparison gives $\Delta e = 0 \iff \Delta O = 0,\ \Delta E = O$. The useful residual is

$$B_u = S_{u-1} + \Delta S_u = \frac{x^{u+1} + (x+1)^{u+1} + 1}{x}.$$

If a, b are even, then $O = S_{u-1}$. Thus $u+1$ is a power of two and $B_u = 0$. The remaining condition is $\Delta(x^{a/2} + x^{b/2}) = 0$, so $a/2, b/2$ are powers of two. These are precisely the even-even pairs in type I.

If a, b are odd, then $E = S_u$ and $B_u = x^{(a-1)/2} + x^{(b-1)/2}$. Its two terms force $\mathrm{wt}_2(u+1) = 2$. Writing $u + 1 = 2^v + 2^w$, $v > w \ge 0$, gives $B_u = x^{2^v - 1} + x^{2^w - 1}$. This yields type III when $w > 0$; $w = 0$ yields type II with $t = 0$.

Finally suppose the missing positions are $2A$ and $2B+1$. Then $O = S_{u-1} + x^B$ and $B_u = x^B + \Delta x^A$. The one-missing-coefficient classification gives either $u = 2^s - 1, B = 0$, or $u = 2^s, B = u - 1$. In the first case $B_u = 0$; in the second $B_u = x^{u-1} + 1$. Both force $\Delta x^A = 1$, so A is a power of two. They give respectively type I with one missing position 1, and type II with $t > 0$. These cases exhaust the parities. All displayed pairs satisfy the required equations on substitution, so the classification is sufficient as well as necessary. □

Theorem 3.3 (Complete four-zero intersection)**.** *For the candidates* (1)*, $\mathcal{M}_n(f_{n,i,j}) = f_{n,i,j}$ if and only if one of the following holds:*

$$n = 2^R + 2^S - 2, \quad R > S \ge 2, \qquad (i, j) = (2^S - 2, 2^S - 1); \tag{9}$$

$$n = 2^R + 2, \quad R \ge 3, \qquad (i, j) = (3, 2^{R-1}); \tag{10}$$

$$n = 14, \qquad (i, j) \in \{(2, 4), (2, 5), (4, 5)\}. \tag{11}$$

Equivalently, these are exactly the four-zero polynomials in the image of $\mathcal{T}$.

Proof. We first use coefficient comparison to reduce the four-zero condition to the two-zero polynomial in Lemma 3.1. We then apply the three types in Lemma 3.2 and impose the second descent identity on each type.

Write $n = 2k$. The coefficient of x^{n-1} in $\mathcal{M}_n(f)$ is $f'(1) = k-2$ in $\mathbb{F}_2$, since the four missing tail positions include two odd positions: each pair $\{r, n-1-r\}$ contains one odd position because $n-1$ is odd. Its value in f is 1, so k is odd. Comparison of the x^2 coefficients gives $[x]f = \binom{2k}{2} = k = 1$. Thus positions $1, n-2$ are not missing. In Lemma 3.1, $e = S_d + x^a + x^b$ is therefore monic of even degree $d = k-1$, with $1 \le a < b \le d-1$.

By Lemma 3.2, e has one of the three types in (7). The second descent condition is equivalent to

$$x^a + x^b + \Delta x^{d+1-a} + \Delta x^{d+1-b} = \frac{x^{d+2} + (x+1)^{d+2} + 1}{x}. \tag{12}$$

For type I the right side is zero. The binary masks $d+1-a$ and $d+1-b$ have all r digits set except the distinct digits specified by a, b. Their proper-submask supports have symmetric difference of size $2^{r-1} - 2$: among all submasks the difference has size 2^{r-1}, and deleting the two top masks removes two terms. Equation (12) requires that difference to be the two terms $x^a + x^b$, so $r = 3$. For $r = 3$ it is exactly that pair, giving $d = 6$ and the three cases (11).

For type II put $\{a, b\} = \{d-1, 2^t\}$. The equation becomes

$$\Delta x^{d+1-2^t} = x^{2^t} + x + 1.$$

For $t = 0$ both sides are 1. For $t \ge 1$ the right side has three terms, while $d+1-2^t$ has $r-t+1$ set digits. Thus $2^{r-t+1} - 1 = 3$, forcing $t = r-1$, which also suffices. The two choices yield respectively (9) with $S = 2$, and (10).

For type III the two Δ terms are both 1 and cancel. The right side is $x^a + x^b$, so every such pair satisfies the condition. Restoring the even zero positions $2a, 2b$ and their tail mirrors gives (9) with $R = r+1, S = s+1 \ge 3$. All these steps are reversible, proving the assertion in every degree. The equivalent image statement follows from the classical representation preceding Lemma 3.1. □

Corollary 3.4 (Reducibility and failure of iteration)**.** *Every member of* (9) *is reducible. If h is itself a four-zero polynomial of the form* (1)*, then $\mathcal{T}(h)$ is never a member of that family. Among the three polynomials in* (11)*, only $f_{14,2,4}$ is irreducible.*

Proof. For $f = \mathcal{T}(h)$ let $m = n/2$; its constant term gives $h(0) = 1$, and the x^2 coefficient gives

$$[x]h = [x^2]f + \binom{m}{2}. \tag{13}$$

In (9), $S = 2$ gives $m \equiv 1 \pmod 4$ and $[x^2]f = 0$, while $S \ge 3$ gives $m \equiv 3 \pmod 4$ and $[x^2]f = 1$. Hence $[x]h = 0$ in either case. If h is reducible, multiplicativity of the transform makes f reducible. If h is irreducible, the classical binary Q criterion [13, Theorem 1.1] says that h^Q, and hence $f = h^Q(x+1)$, is irreducible exactly when $[x]h = 1$. This proves the first assertion.

The theorem, or its coefficient proof, gives $n \equiv 2 \pmod 4$ for every four-zero output. A four-zero input has even degree, so its transformed degree is divisible by four. This proves the iteration obstruction for this particular transform.

For the exceptional pairs $(2,4), (2,5), (4,5)$ the corresponding h are, respectively,

$$x^7 + x^6 + x^3 + x + 1, \quad x^7 + x^6 + x^3 + x^2 + x + 1, \quad x^7 + x^6 + x^4 + x^2 + 1.$$

The first satisfies $x^{128} \equiv x \pmod{h}$ and $\gcd(h, x^2 + x) = 1$, so is irreducible by the prime-degree Rabin test; its linear coefficient is 1. The second has root 1, and the third has zero linear coefficient. The same Q criterion finishes the proof. □

Proposition 3.5 (Explicit descent for the B family)**.** *For $R \geq 3$, put $q = 2^{R-1}$. The unique polynomial with $\mathcal{T}(h_R) = f_{2q+2,3,q}$ is*

$$h_R(x) = 1 + x + x^2 + (1 + x)\left(x^q + \sum_{t=2}^{R-2} x^{q-2^t+1}\right), \tag{14}$$

where the sum is empty for $R = 3$. Moreover, $\deg h_R = q + 1$ and $[x]h_R = 1$.

Proof. Set $L(x) = \sum_{a=0}^{R-2} x^{2^a}$ and

$$F(x) = x^{q+2} h_R(1/x) = (x^2 + x + 1)x^q + (x + 1)(L(x) + x^2).$$

For $s = X^{-1}$ and $w = s + s^2$, telescoping gives $L(w) = s + s^q$; substitution and collection of terms give

$$s^{-2q-4} F(w) = (X + 1) f_{2q+2,3,q}(X).$$

The left side is $(X+1)\mathcal{T}(h_R)(X)$, proving the formula; the distinct leading degrees of the transform basis give uniqueness. The exponents in the sum lie between $q/2 + 1$ and $q - 3$, so the degree and linear coefficient follow directly. □

The classical Q criterion makes irreducibility of h_R equivalent to that of its output.

3.2 The translation branch

We next impose $f(x+1) = f(x)$. The invariant representation $f = H(x^2+x)$ and its Artin–Schreier irreducibility criterion are classical [12, Lemma 2.1 and Theorem 2.2]. The additional constraints below come from the tail palindrome and its missing coefficients.

Lemma 3.6 (Degree compatibility)**.** *If a monic tail-palindromic polynomial f of even degree $n \geq 6$ satisfies $f(x + 1) = f(x)$, then $n + 2$ is a power of two.*

Proof. Let $F(X, Y) = Y^n f(X/Y)$ and $N = n + 1$. Tail symmetry gives

$$XF(X, Y) + YF(Y, X) = X^N + Y^N.$$

By translation invariance, put $A = F(X, Y) = F(X + Y, Y)$, $B = F(Y, X) = F(X + Y, X)$ and $C = F(X, X + Y) = F(Y, X + Y)$. Applying the displayed identity to three pairs of variables yields

$$\begin{aligned} XA + YB &= X^N + Y^N, \\ (X + Y)A + YC &= (X + Y)^N + Y^N, \\ XC + (X + Y)B &= X^N + (X + Y)^N. \end{aligned}$$

Multiply these equations by $X + Y$, X and Y, respectively, and add. All terms on the left cancel, leaving $(X + Y)^{n+2} = X^{n+2} + Y^{n+2}$. In characteristic two this forces $n + 2$ to be a power of two: if its binary expansion has at least two nonzero digits, a singleton proper submask gives a nonzero mixed binomial term. □

Proposition 3.7 (A missing-coefficient obstruction)**.** *Let f be monic of even degree $n = 2k \geq 6$, with constant term one and a tail palindrome containing $2s$ zeros. If $f(x+1) = f(x)$, write $f = H(x^2 + x)$ with H monic of degree k. Then*

$$[x^{k-1}]H = s \pmod 2. \tag{15}$$

In particular, if the number of tail zeros is divisible by four, then f is reducible.

Proof. With e, o as in Lemma 3.1, tail symmetry gives $f = (x^k+e)^2+xo^2$. Odd and even coefficient comparison gives

$$\Delta f = 0 \quad \Longleftrightarrow \quad \Delta o = 0, \qquad \Delta(x^k + e) = o. \tag{16}$$

Indeed $\Delta f = x(\Delta o)^2 + [\Delta(x^k + e) + o(x+1)]^2$. The formal coefficient vector of o has length k and s zeros, so $[x^{n-2}]f = o(0) = o(1) = k - s$ in $\mathbb{F}_2$. Lemma 3.6 gives $k = 2^{R-1} - 1 \equiv 3 \pmod 4$, where $R \geq 3$. The classical coefficient comparison in $H(x^2 + x)$ therefore gives

$$[x^{k-1}]H = [x^{2k-2}]f + \binom{k}{2} = s \pmod 2.$$

Now suppose that s is even. If H is reducible, so is its composition. Otherwise let β be a root of the irreducible H in $\mathbb{F}_{2^k}$. Its absolute trace is $[x^{k-1}]H = 0$, so $z^2 + z = \beta$ has a solution in that same field. Thus f has a root of degree at most k and is reducible. The last step is classical: specialize [11, Theorem 2.2(2)] to $p = 2$, $n = r = 1$ and $a = 1$ in Agou's notation, with irreducible outer polynomial H; see also [12, Theorem 2.2]. The same binary specialization follows from Long [10, Theorems 4.1–4.2, pp. 69 and 71]; the basic Artin–Schreier trace criterion is classical, as noted in [10, p. 63]. The associated high-coefficient restriction appears in [12, Theorem 2.8(3)]. □

The restriction $n \geq 6$ is essential: x^2+x+1 is a translation-invariant irreducible tail palindrome with no zeros. For odd s, (15) only removes this trace obstruction; it does not establish irreducibility of H.

Theorem 3.8 (Complete translation intersection)**.** *The polynomial $f_{n,i,j}$ in* (1) *satisfies $f_{n,i,j}(x+1) = f_{n,i,j}(x)$ if and only if*

$$(n, i, j) \in \{(14, 3, 4), (30, 5, 9), (30, 5, 12), (30, 9, 12)\}. \tag{17}$$

All four polynomials are reducible.

Proof. In (16), put $d = k - 1$. The polynomial o is monic of degree d and has two missing coefficients. Since k is odd, $o(1) = k - 2 = 1$; invariance gives $o(0) = 1$. Thus $o = S_d + x^a + x^b$ with $1 \leq a < b \leq d - 1$, and $e = S_d + x^{d-a} + x^{d-b}$. By Lemma 3.6, $d = 2^r - 2$, $r \geq 2$. The case $r = 2$ has only one interior position and is impossible. For $r \geq 3$, Equation (8) gives $\Delta S_d = 0$. Hence $\Delta x^a = \Delta x^b$, which, by the binary-support observation in the proof of Lemma 3.2, forces a, b to be distinct powers of two. The second condition in (16), using $\Delta x^{d+1} = S_d$, becomes

$$\Delta x^{d-a} + \Delta x^{d-b} = x^a + x^b. \tag{18}$$

If a, b are both even, take square roots in this equation. The two exponents on the left are $(r-1)$-digit all-one binary masks with different singleton digits removed. Their full submask supports have symmetric difference of size 2^{r-2}; removing the two top masks leaves $2^{r-2} - 2$ terms. The right side has two terms, so $r = 4$. For three-digit masks the remaining difference is precisely the two removed singleton digits; thus all pairs from $\{2, 4, 8\}$ satisfy (18).

Otherwise $a = 1$ and b is an even power of two. The term Δx^{d-b} has only even exponents. The mask $d - 1 = 2^r - 3$ has all digits except digit one set, so Δx^{d-1} has $2^{r-2} - 1$ odd terms. The right side has only the odd term x, forcing $r = 3$. Here $\Delta x^5 = x^4 + x + 1$ and $\Delta x^4 = \Delta x^2 = 1$; of $b = 2, 4$, exactly $b = 4$ satisfies the equation.

A missing position a in o corresponds to the two positions $2a + 1$ and $n - 2 - 2a$ in f. Thus $d = 6$, $\{a, b\} = \{1, 4\}$ gives $(14, 3, 4)$; $d = 14$ and the three pairs from $\{2, 4, 8\}$ give the degree-30 cases in (17). The substitutions above also prove sufficiency. Reducibility follows uniformly from Proposition 3.7 with $s = 2$. □

3.3 Full-degree quadratic constructions

Here a full-degree quadratic pullback over $\mathbb{F}_2$ means

$$F = h^m P(g/h), \qquad \gcd(g, h) = 1, \qquad \max(\deg g, \deg h) = 2,$$

where $\deg P = m > 1$ and $\deg F = 2m$. The last condition is automatic for irreducible P; in general the degree drops exactly when $h_2 \neq 0$ and $P(g_2/h_2) = 0$ [8, Section 2].

Corollary 3.9 (Boundary for full-degree quadratic pullbacks)**.** *If a full-degree quadratic pullback is a four-zero tail palindrome, then it belongs either to the lists in Theorem 3.3 or to* (17)*, and its degree is* 2 *modulo* 4*. If it is irreducible, it is either* $f_{14,2,4}$ *or a member of* (10)*. Consequently, no sequence of quadratic pullbacks, each doubling the degree, can reach a four-zero tail palindrome after starting at an even degree.*

Proof. Write $g = \sum_{j=0}^{2} g_j x^j$, $h = \sum_{j=0}^{2} h_j x^j$, and put

$$a = g_2 h_1 + g_1 h_2, \quad b = g_0 h_2 + g_2 h_0, \quad c = g_1 h_0 + g_0 h_1.$$

The classical quadratic framework gives $b^2 + ac = 1$ and, in the separable case,

$$(ax + b)^{2m} F\left(\frac{bx + c}{ax + b}\right) = F(x) \qquad \text{[8, Section 4, Lemma 9]}.$$

In the separable case the possible triples are exactly $(1, 0, 1), (0, 1, 1), (1, 1, 0)$, corresponding to reciprocity, translation and $\mathcal{M}_{2m}$. There is no nontrivial scalar normalization over $\mathbb{F}_2$. In the inseparable case $g_1 = h_1 = 0$, the output is a square; its odd coefficients vanish, whereas a four-zero candidate has $[x^{2m-1}]F = 1$.

The reciprocal branch contains no four-zero candidate: combining $f^* = f$ with Lemma 2.1 gives $(x + 1)f = x^{n+1} + 1 = (x + 1)S_n$, hence $f = S_n$. The other two branches are classified by Theorems 3.3 and 3.8. The irreducible alternatives follow from their reducibility conclusions. All listed degrees are 2 modulo 4. Starting at an even degree, every subsequent doubled degree is divisible by four, proving the last assertion. □

The three involutions thus connect the support symmetries to an exhaustive list of full-degree quadratic outputs.

4 Primitive counts and reproducible computations

Complete enumeration of the 161700 candidates at the 98 even degrees $6 \leq n \leq 200$ gives

$$\sum_{\substack{6 \leq n \leq 200 \\ n \text{ even}}} N_4(n) = 2663, \qquad N_4(n) > 0 \text{ at every tested degree.}$$

A separate implementation reproduces every complete count and first pair. At the 50 even degrees $202 \le n \le 300$, checking the lexicographic prefixes covers 5410 further candidates and verifies the first pair at each degree.

n	candidates	$N_4(n)$	first (i, j)
6	1	1	$(1, 2)$
8	3	1	$(2, 3)$
10	6	2	$(1, 2)$
12	10	3	$(1, 3)$
20	36	1	$(4, 5)$
40	171	11	$(1, 4)$
62	435	16	$(1, 17)$
100	1176	31	$(1, 23)$
126	1891	55	$(1, 2)$
154	2850	45	$(1, 23)$
194	4560	20	$(6, 65)$
200	4851	17	$(1, 71)$

Table 1: Selected complete counts. All 98 rows and all successful position pairs through degree 200 are supplied with the verification data.

Arithmetic verification and reproducibility. The verification code constructs coefficients directly and tests irreducibility, followed by the standard exact-order conditions

$$x^{2^n-1} \equiv 1 \pmod{f}, \qquad x^{(2^n-1)/p} \not\equiv 1 \pmod{f} \quad (p \mid 2^n - 1 \text{ prime}). \tag{19}$$

The arithmetic tests use python-flint 0.9.0 with FLINT 3.6.0 [14]. The 148 order factorizations correspond to the 98 complete-count degrees and the 50 further first-witness degrees. Exact division leaves residual cofactor one; all 484 distinct prime factors, of at most 144 bits, pass FLINT's rigorous `fmpz_is_prime` procedure. A standard-library implementation independently agrees on all 220 candidates through degree 24.

The folder `verification/primitivity/` supplies complete candidate verdicts, successful positions, exact order-test residues for the 148 first witnesses, recovered integer factorizations and input hashes. Historical inputs and correction records are preserved with the audit. The data and verification programs are publicly archived at doi:10.5281/zenodo.23766722 and in the fixed GitHub release.

The companion programs also check the reciprocal filter, both quadratic classifications and the descent identities; detailed ranges accompany the archived outputs. The four translation outputs have independently certified factor degrees $2+12$, $12+18$, $6+24$ and $7+7+8+8$, with a repeated factor in the last. The B-sequence factorization data cover $3 \le R \le 10$: the first term is irreducible and the next seven reducible.

5 Structural consequences and open questions

The reciprocal screen counts positions compatible with reciprocal coprimality. The quadratic classifications determine every full-degree output and yield a uniform obstruction from translation symmetry.

Irreducibility in the remaining branch. Within full-degree quadratic constructions, the B family (10) is the remaining irreducibility question in Corollary 3.9. Which polynomials h_R of Proposition 3.5 are irreducible? The explicit descent and the binary Q criterion make this equivalent to classifying the irreducible four-zero outputs in that branch.

Primitive existence and further constructions. Conjecture 1.1 asks for primitive members at every even degree, linking the prescribed support to the multiplicative order of roots. The complete counts and first witnesses give concrete instances; the quotient and finite-field descriptions in Appendix A express the support condition and the order condition explicitly. Higher-degree maps, selected factors and degree-dropping substitutions provide further construction questions.

A Coefficient and finite-field descriptions

The coefficient quotient organizes position pairs into classes with explicit multiplicities; the finite-field parametrization encodes primitivity through coprime order components.

A.1 Coefficient quotient

Let ρ reverse $\mathbb{F}_2^k$, and set

$$P = \{(A, \rho A) : A \in \mathbb{F}_2^k\}, \quad K = \{(C, C) : C = \rho C\}, \quad U = P/K, \quad r = \lfloor k/2 \rfloor.$$

The map $[(A, \rho A)] \mapsto A + \rho A$ identifies U with $\mathrm{im}(I + \rho)$, of dimension r. Prepending a zero to $x \in \mathbb{F}_2^{k-1}$ defines $\Phi(x) = [(\widetilde{x}, \rho\widetilde{x})]$. The all-ones tail lies in K, so deleting positions i, j gives class $\Phi(e_i + e_j)$. With zero-based coordinates and a suitable basis $z_1, \ldots, z_r$, the columns $u_i = \Phi(e_i)$ form the multiset

$$\{u_1, \ldots, u_{k-1}\} = \begin{cases} \{z_1, z_1, \ldots, z_{r-1}, z_{r-1}, z_r\}, & k = 2r, \\ \{z_1, z_1, \ldots, z_{r-1}, z_{r-1}, z_r, 0\}, & k = 2r + 1. \end{cases} \tag{20}$$

Indeed pair positions $i, k - 1 - i$; the boundary pair contributes the singleton z_r and an unpaired middle position contributes zero. In particular Φ is onto.

For $s = \sum s_a z_a$, write $h = \sum_{a<r} s_a$ as an integer and $\varepsilon = k \bmod 2$. The ordinary coset weight enumerator is

$$A_s(z) = \sum_{\Phi(x)=s} z^{\mathrm{wt}(x)} = (1 + z)^{\varepsilon} z^{s_r} (2z)^h (1 + z^2)^{r-1-h}. \tag{21}$$

Each repeated column allows $00, 11$ for zero syndrome and $01, 10$ for nonzero syndrome; the singleton and optional zero column give the remaining factors. Thus $\mu(s) = [z^2]A_s(z)$ counts four-zero candidates in class s, and the number of occupied classes is

$$D(3) = 1, \qquad D(k) = 1 + \binom{\lfloor (k+1)/2 \rfloor}{2} \quad (k \geq 4). \tag{22}$$

This follows by counting the nonzero quadratic coefficients in (21); at $k = 3$ the zero class is unoccupied. For a character χ of U, put $t = \#\{i : \chi(u_i) = -1\}$ and $\ell = k - 1$. Then

$$\widehat{\mu}(\chi) = \sum_{i<j} \chi(u_i)\chi(u_j) = K_2(t; \ell) = \frac{(\ell - 2t)^2 - \ell}{2}. \tag{23}$$

These are classical direct-product weight-enumerator and Krawtchouk identities [7, Sections 3–4][6, Sections 1.1 and 2.1]. The different factorization behavior within a class is visible already at degree eight: $x^8+x^7+x^6+x+1$ is primitive, while $x^8+x^7+x^5+x^2+1=(x^2+x+1)(x^6+x^4+x^2+x+1)$ is reducible; their tail difference 0110 0110 belongs to K.

A.2 Finite-field parametrization

Put $q=2^k$, $\mathbb{K}=\mathbb{F}_q$, $\mathbb{L}=\mathbb{F}_{q^2}$ and $T=\{\gamma\in\mathbb{L}^*:\gamma^{q+1}=1\}$. The coprime orders $q-1, q+1$ give $\mathbb{L}^*=\mathbb{K}^*\times T$. A primitive element is uniquely $\beta=\nu\gamma$ with $\mathrm{ord}(\nu)=q-1$ and $\mathrm{ord}(\gamma)=q+1$. Writing $\tau=\gamma+\gamma^{-1}\in\mathbb{K}$, the standard Dickson identity $D_a(u+u^{-1})=u^a+u^{-a}$ and Frobenius pairing give

$$F_{\nu,\gamma}(x)=\prod_{i=0}^{k-1}\left(x^2+\nu^{2^i}D_{2^i}(\tau)x+\nu^{2^{i+1}}\right). \tag{24}$$

Here $D_0=0, D_1=t, D_{a+1}=tD_a+D_{a-1}$ in characteristic two [2]. Conversely, the two order conditions make this product the minimal polynomial of a primitive element. Expanding the product and comparing with (1) gives an exact coefficient-and-order system in which the support and the two exact-order conditions are imposed simultaneously.

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